

Decoding 5G Telemetry

Reverse Engineering Qualcomm
0xB881 Logs for MAC Uplink
Performance Analysis



The Challenge

- **Objective:** Decrypt Qualcomm 0xB881 (MAC Uplink TB Statistics) logs to accurately measure physical layer performance.
 - **Data:** Working entirely with raw, undocumented hexadecimal byte streams extracted from modem diagnostics.
 - **Work Done:**
 - We develop a custom parser to map these byte sequences into structured, network-level metrics based on insights gained from the MobileInsight codebase.
 - Then, we test the veracity of these parsed metrics by running them on real-world traces, tracked over different network circumstances, and looking at the patterns in the MAC level data and comparing them with application layer outputs.
-



Methodology

- **Baseline:** MobileInsight's previous-gen C parser(v2.0/2.1) used to identify standard byte offsets, timestamp structures, and bitmasks.
 - **Extrapolation:** To map the new 80-byte fixed-length transport block record structure(v3.1).
 - **Post-processing:** Sequential processing in Python to track cumulative counters (New Tx, ReTx, RBs, and MCS) across bins, yielding precise throughput and physical resource allocations.
-

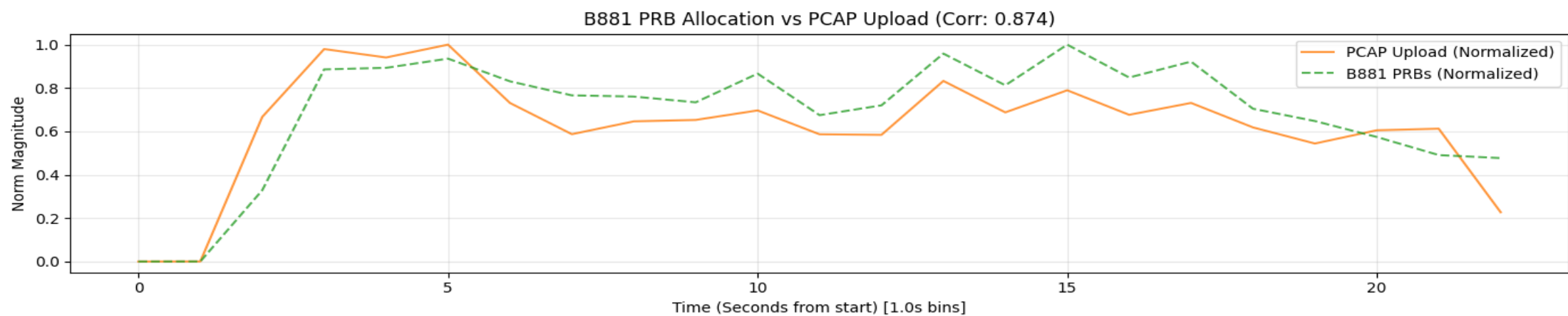
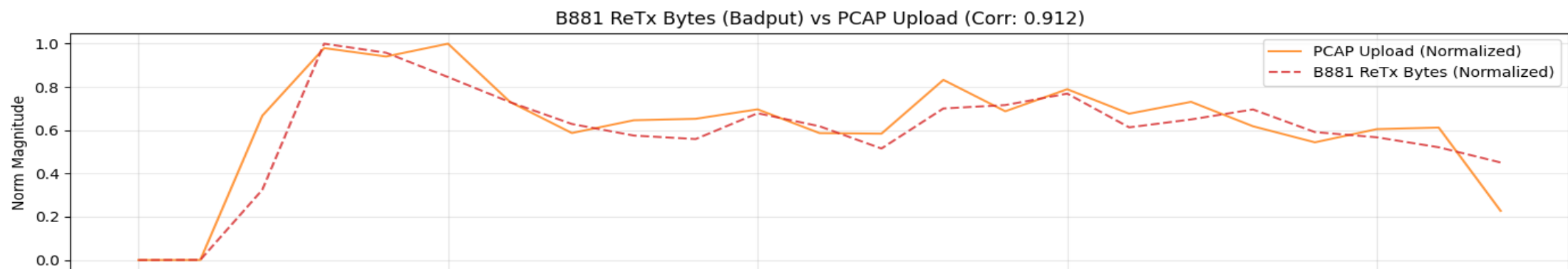
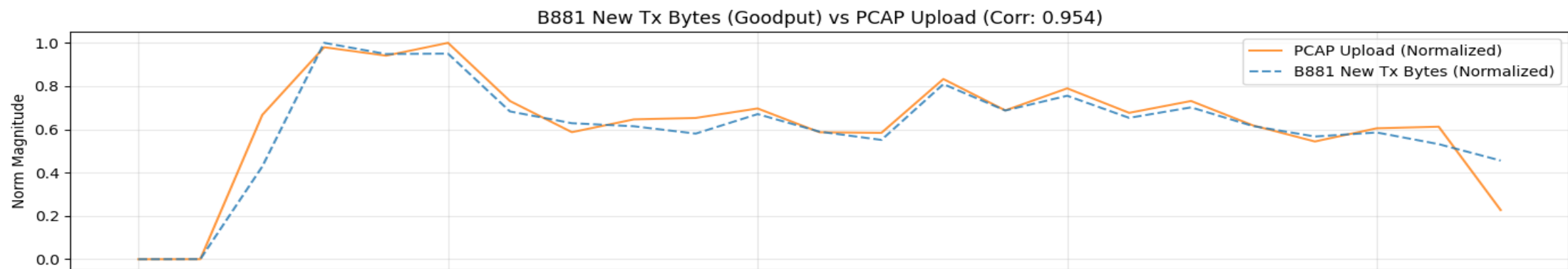
Parsed 0xB881 (v3.1) Structure

Field Name	Byte Offset	Size	Metric Description
New Tx Bytes	O + 0	8 Bytes	Cumulative raw payload volume (Goodput).
ReTx Bytes	O + 8	8 Bytes	Cumulative retransmitted payload volume (Badput).
Num MCS	O + 16	8 Bytes	Sum of Modulation and Coding Scheme indices.
Num PRB	O + 24	8 Bytes	Physical Resource Blocks allocated by the base station.



PCAP Traffic Correlation (Cross-Layer Validation)

- To verify the accuracy of our baseband extraction, we aligned the MAC-layer telemetry with network-layer packets captured via tcpdump.
 - By resolving timestamp offsets and aggregating data into 1.0-second bins, we calculated the correlation between MAC transmission deltas and actual IP upload traffic.
 - **Result:** Typical runs showed correlation coefficients $r > 0.95$, proving the mapped MAC deltas accurately reflect net data usage.
 - While this was true for Goodput, Badput, and Physical Resource Blocks, we still needed a way to confirm whether the MCS values we collected were indeed accurate. To that end we performed two additional experiments.
-





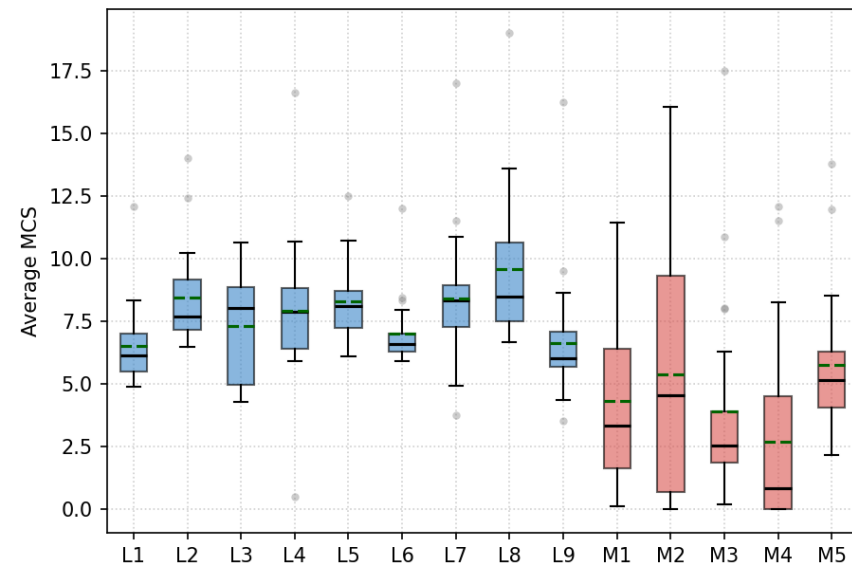
Experiment 1: Stationary vs. Moving Lab

- **Objective:** To benchmark device uplink performance in a stationary environment(L1-L9) against a dynamic environment (M1-M5).
 - Stationary Lab Baseline
 - In a controlled, stationary RF environment (L1-L9), the 5G connection achieved a mean uplink throughput of **15.53 MBps**.
 - The network maintained a high mean Modulation and Coding Scheme (MCS) of **7.79** while keeping the Block Error Rate (BLER) around a steady **8.96%**.
 - Link Adaptation in Motion
 - Under simulated mobility (M1-M5), fast-fading channel states degraded performance. Throughput dropped by **57.4%** to just **6.61 MBps**.
 - The average MCS was **4.46** and the BLER (Block Error Rate) was **9.42%**.
-

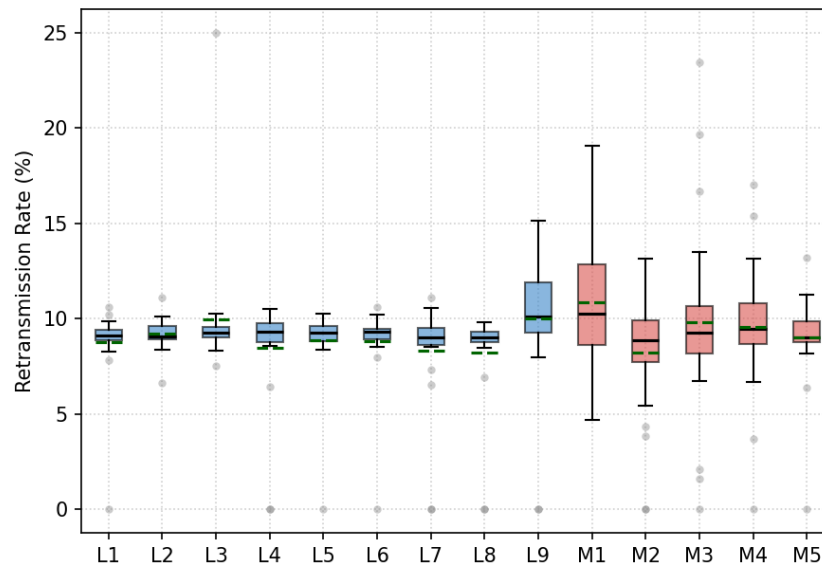
Uplink Performance Comparison: Stationary Lab vs Moving Lab

Stationary Lab (L1-L9)
Moving Lab (M1-M5)

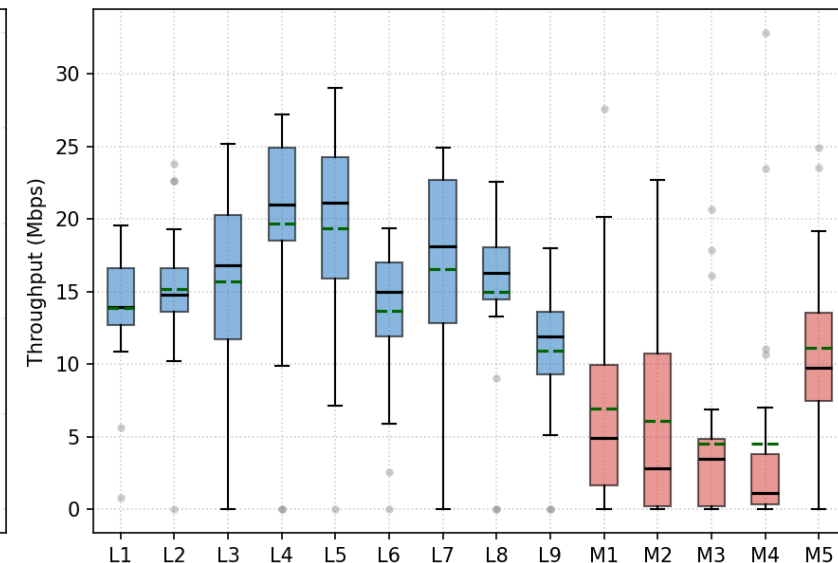
MCS Index per Run



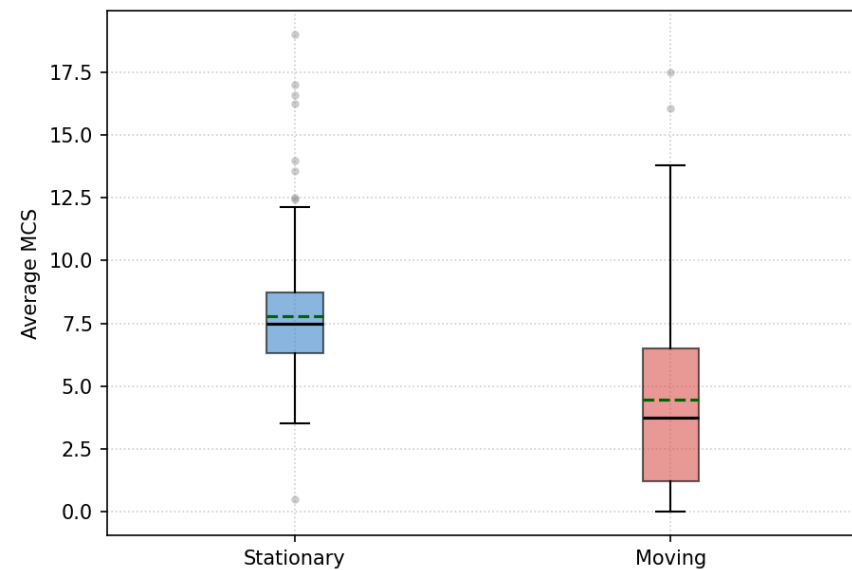
Uplink BLER per Run



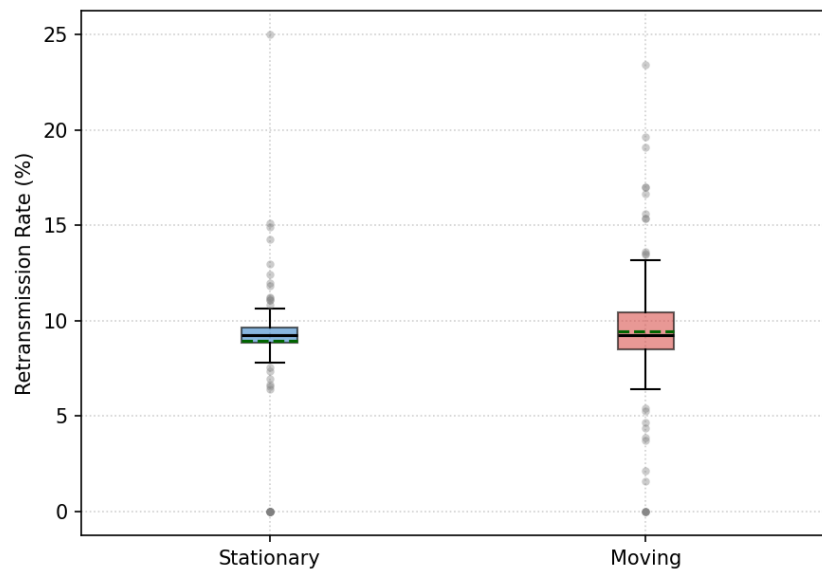
Uplink Throughput per Run



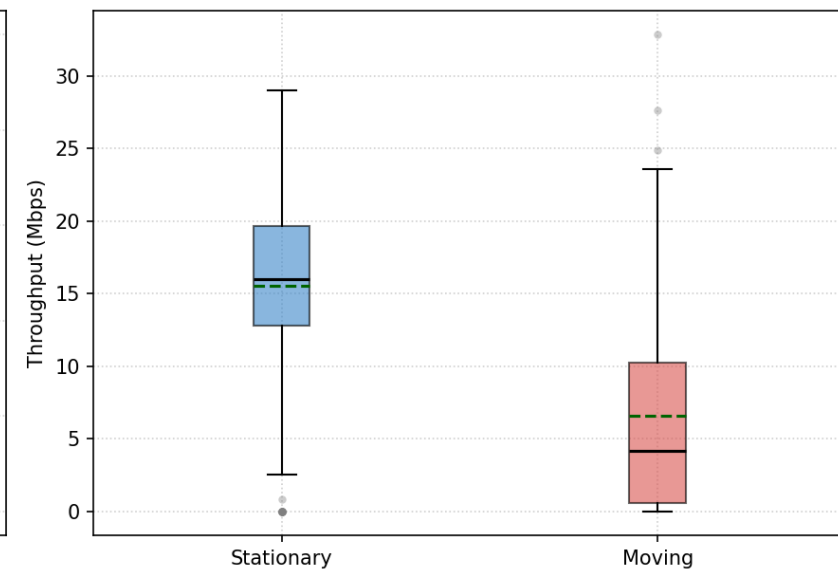
Aggregated MCS Index Comparison



Aggregated Uplink BLER Comparison



Aggregated Uplink Throughput Comparison

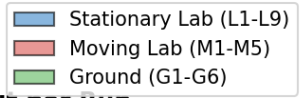




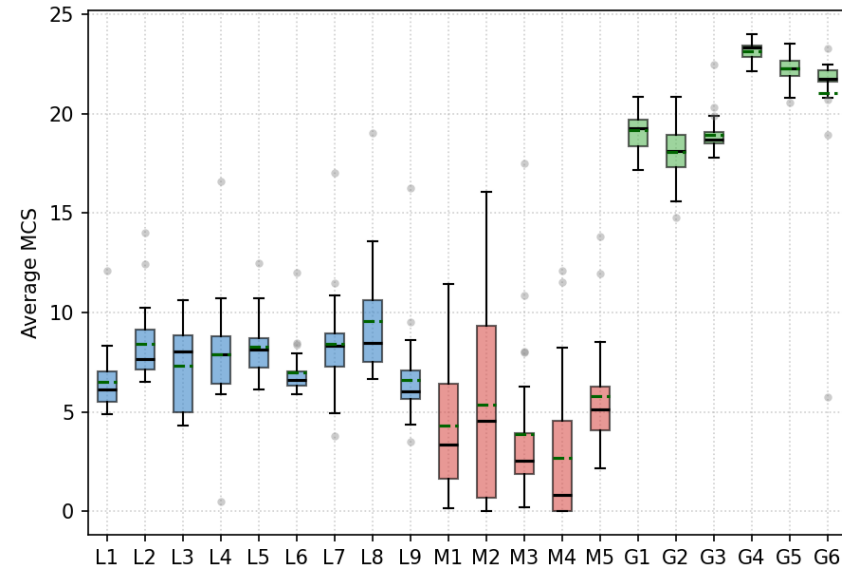
Experiment 2: Lab vs. Real-World Ground

- **Objective:** To benchmark device uplink performance in a closed room environment(lab)(L1-L6) against an open field(ground) environment (G1-G6).
 - **Results:** The lab averaged an uplink throughput of 16.24 MBps. Whereas the ground deployment—benefiting from a higher RSRP environment—delivered a mean throughput of 58.17 MBps, representing a 3.6x improvement in speed
 - This surge in throughput was directly driven by the base station successfully scheduling a significantly higher average Modulation and Coding Scheme (MCS of 20.43 vs. 7.57)
-

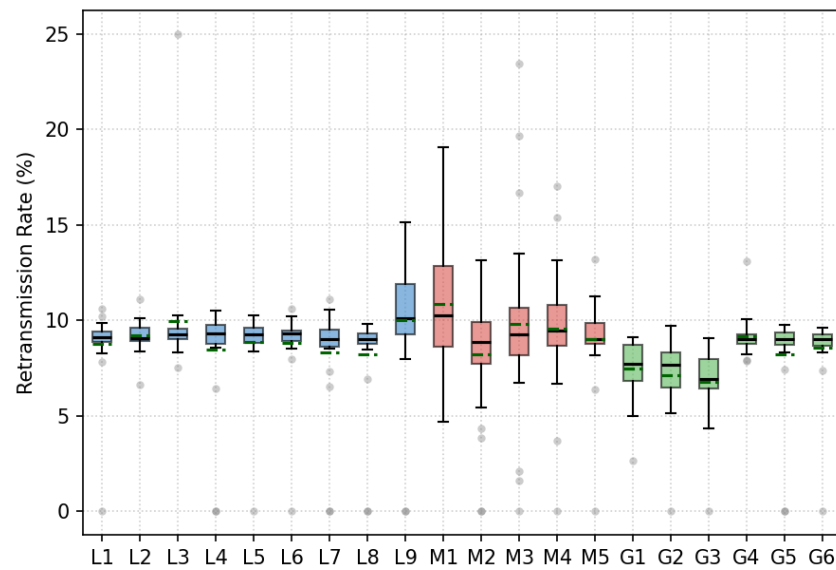
Uplink Performance Comparison: Lab vs Moving vs Ground



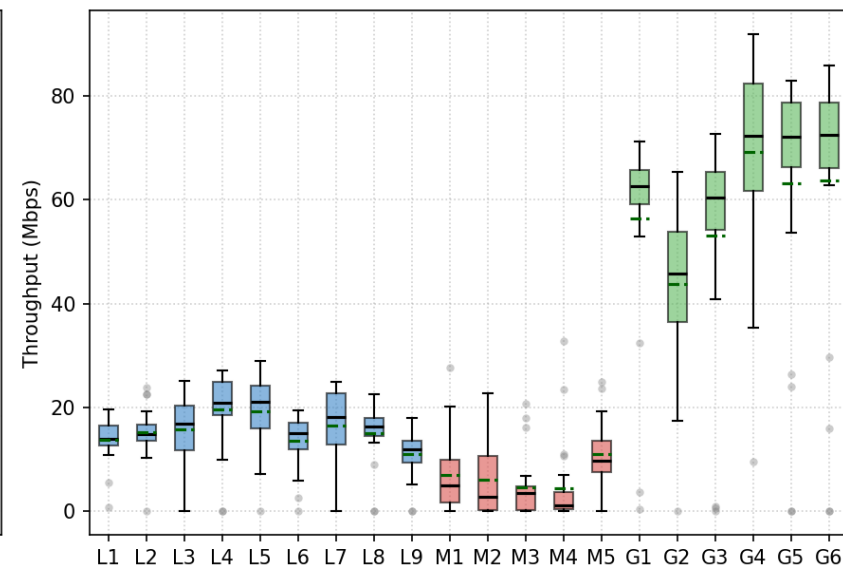
MCS Index per Run



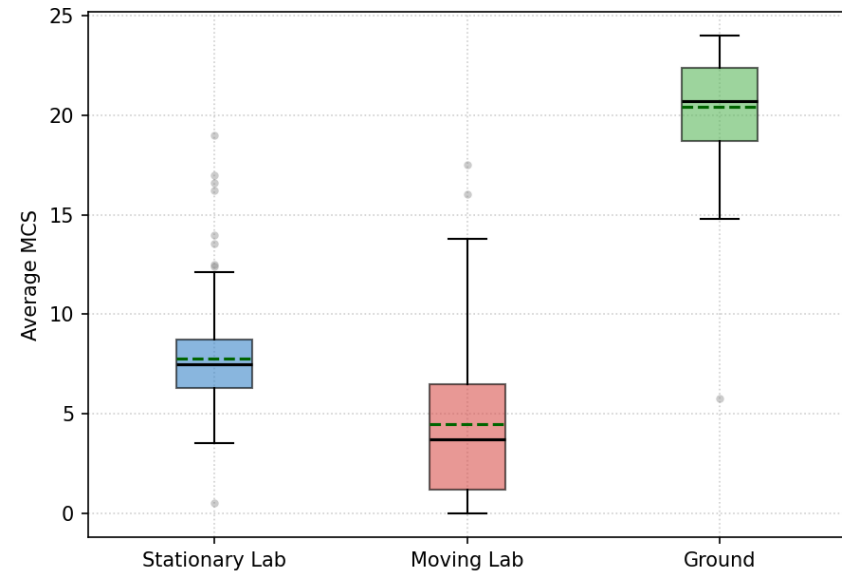
Uplink BLER per Run



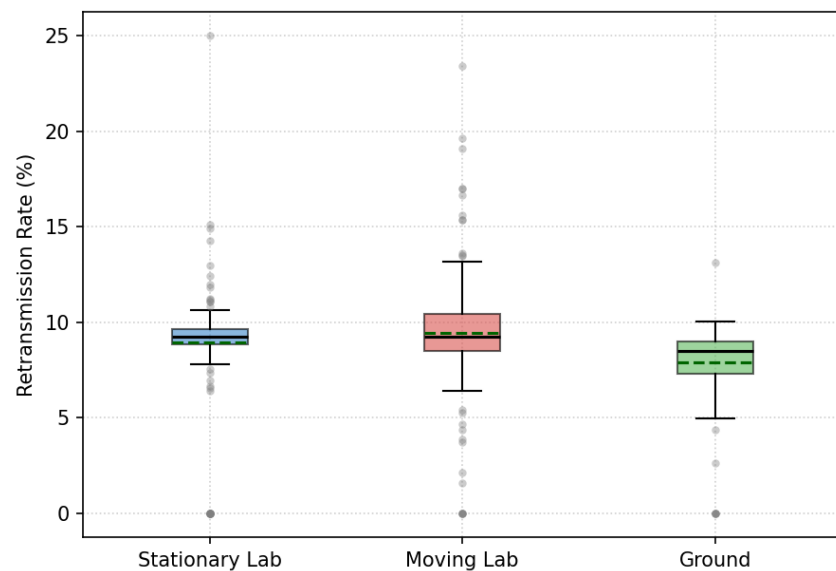
Uplink Throughput per Run



Aggregated MCS Index Comparison



Aggregated Uplink BLER Comparison



Aggregated Uplink Throughput Comparison

